Global Developer Technology Trends Analysis: A Data-Driven Capstone Project

Joseph Junior Donkoh

IBM Data Analyst Professional Certificate, Project Report

February 2026

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion

EXECUTIVE SUMMARY

- This project analyzes developer survey data to identify current technology usage, emerging technology trends, and demographic patterns within the global developer community.
- Key findings:
- Full-stack combinations (HTML/CSS/JavaScript/TypeScript) dominate current usage.
- PostgreSQL leads both current database usage and future demand.
- Cloud platforms, particularly AWS and Azure, show strong growth trends.
- The majority of respondents are aged 25–34 with bachelor's degrees.
- The analysis integrates data cleaning in Jupyter Notebook and interactive visualization using IBM Cognos Analytics, providing actionable insights for developers, employers, and technology decision-makers.

INTRODUCTION

- **Purpose**

To analyze global developer technology usage and identify future trends.

Target Audience

- Hiring managers
- Technology strategists
- Developers planning career paths
- Industry analysts

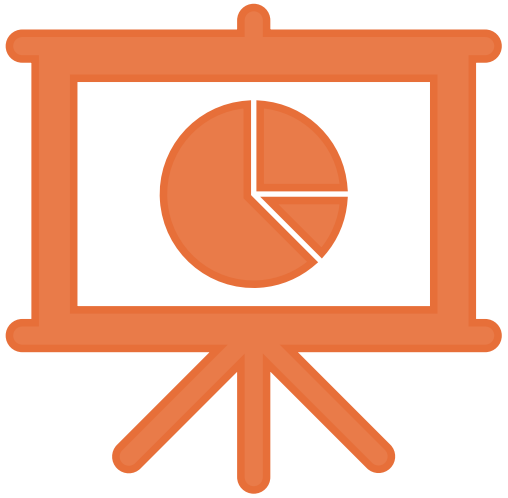
Value of Analysis

- Supports workforce planning
- Identifies high-demand skills
- Reveals emerging technology shifts
- Provides demographic insights

METHODOLOGY

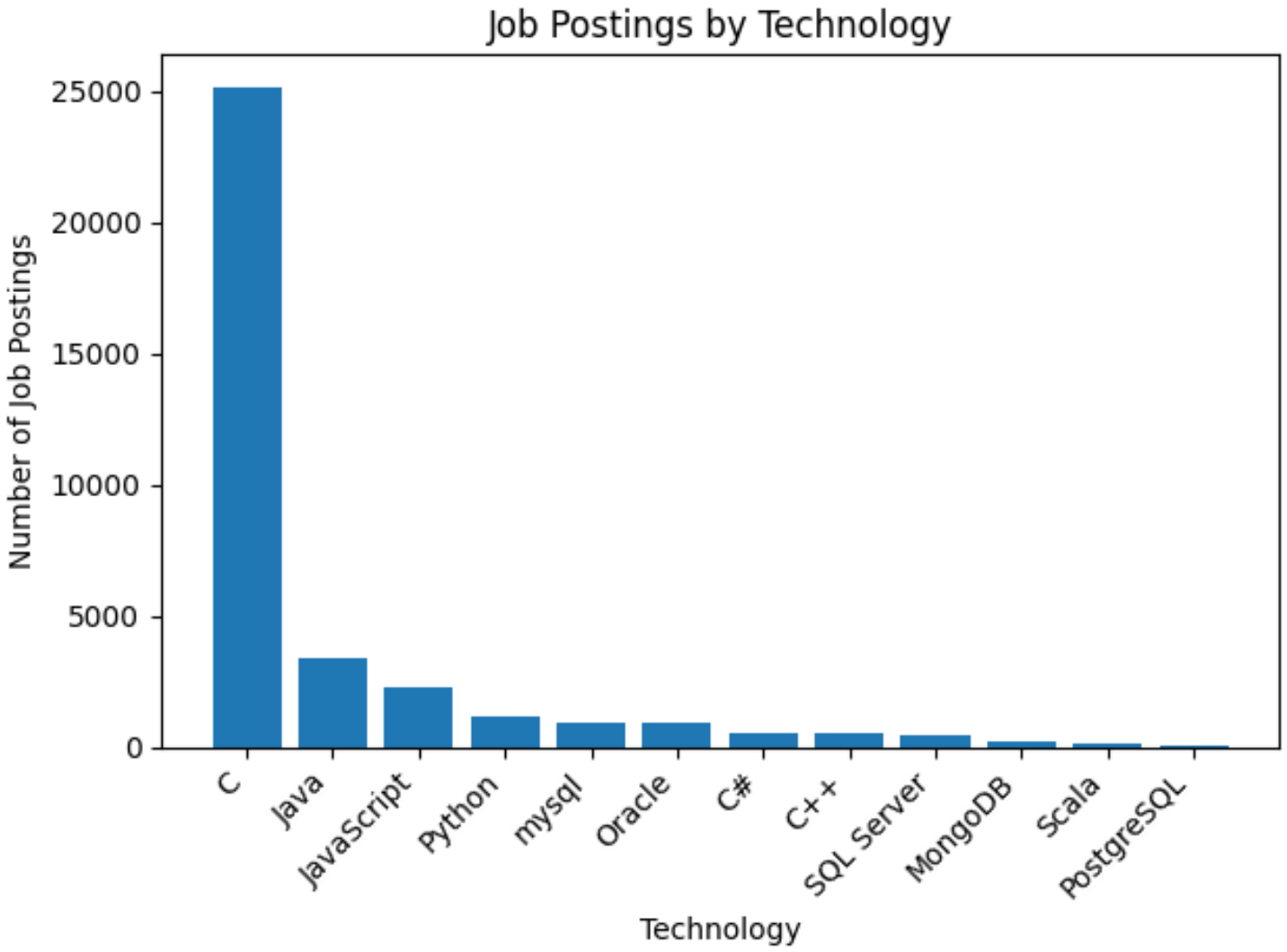
- **Data Sources**
- Developer survey dataset (survey_data_updated.csv)
- Cleaned and analyzed using Jupyter Notebook
- Visualized using IBM Cognos Analytics
- **Data Wrangling Steps**
- Removed duplicates
- Handled missing values
- Converted data types
- Applied aggregation (Count)
- Created Top 10 filters
- Conducted exploratory data analysis
- **Tools Used**
- Python (Pandas, NumPy)
- Matplotlib & Seaborn
- IBM Cognos Analytics

Results

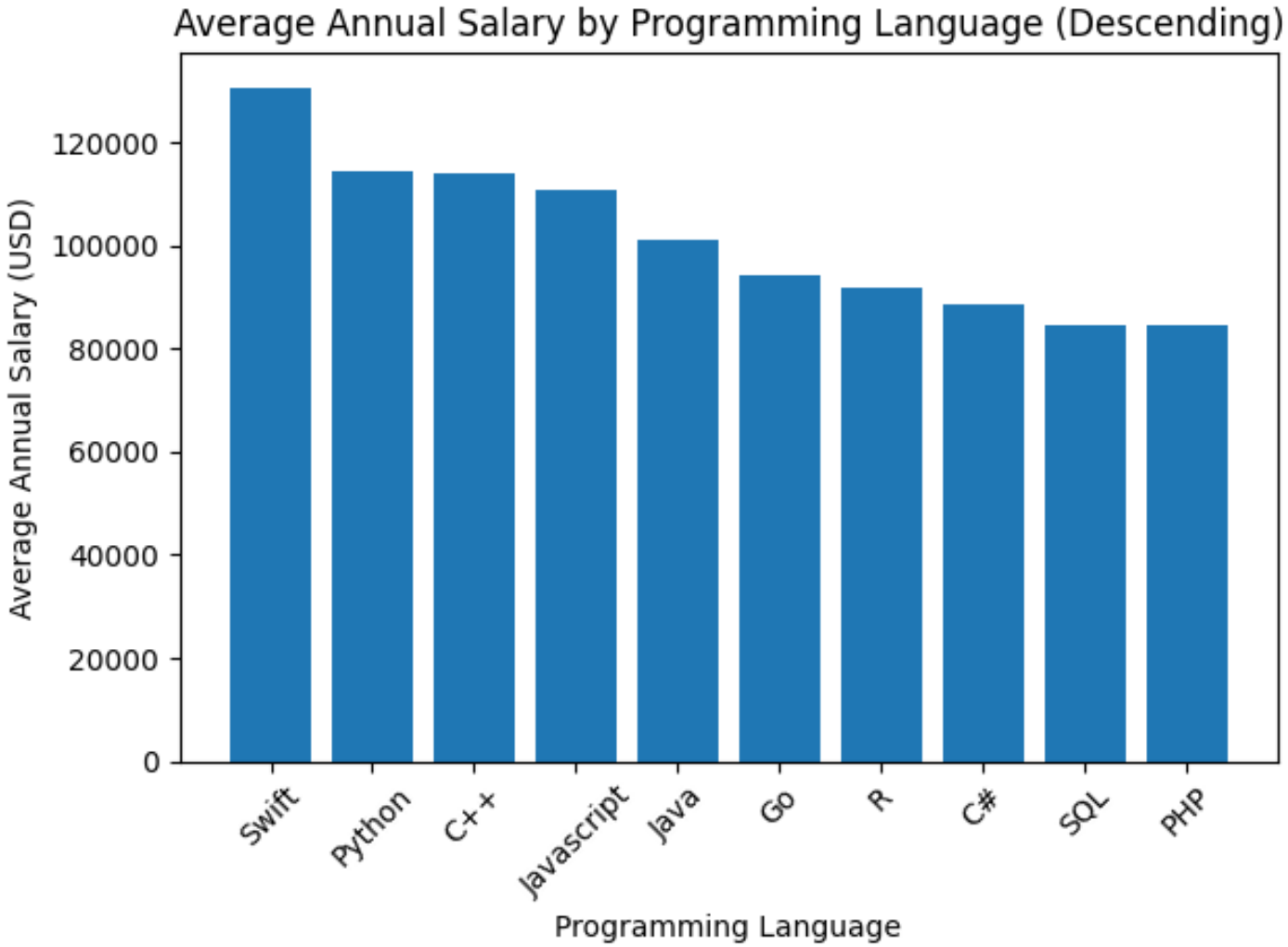


>>>

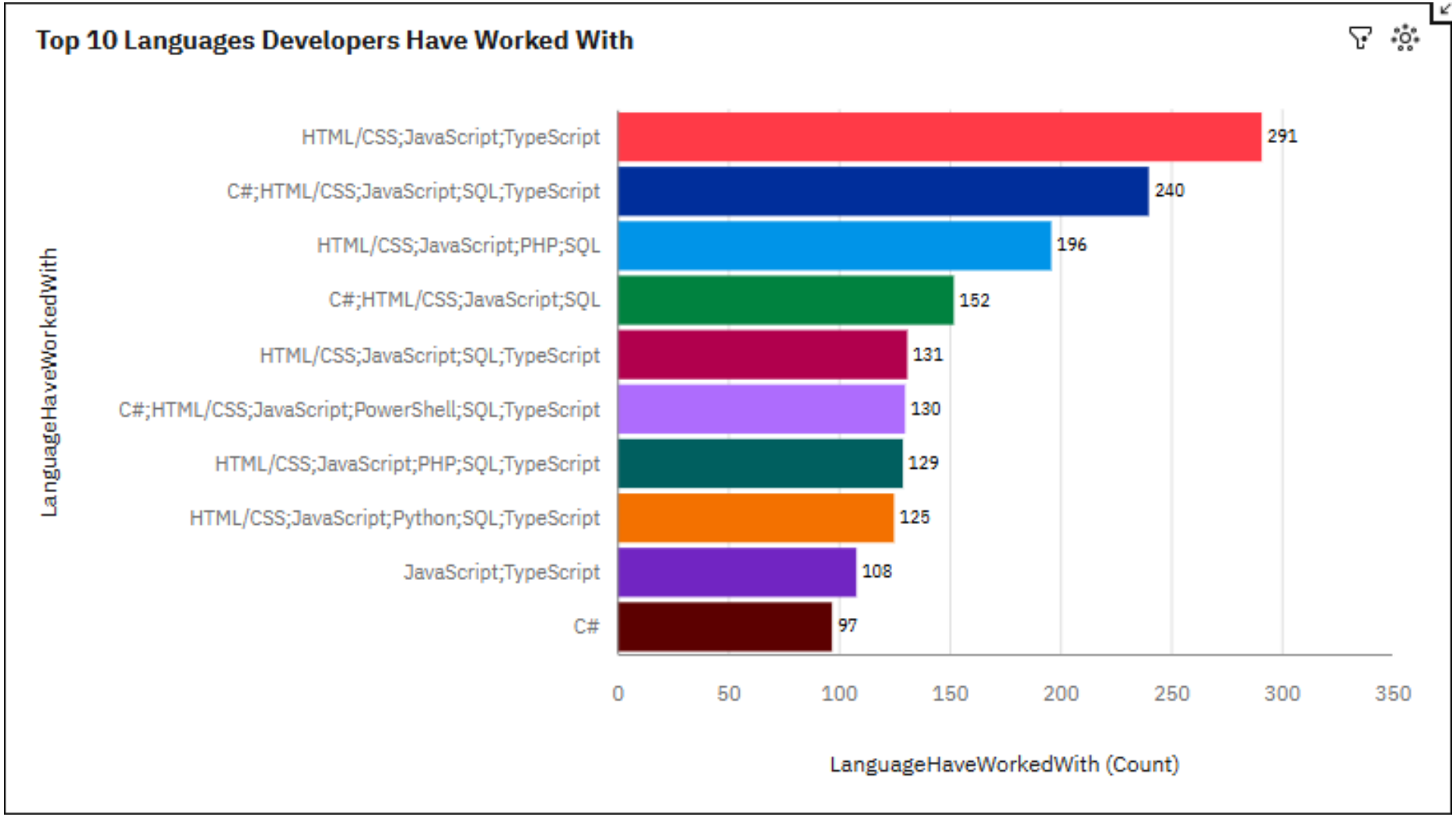
JOB POSTINGS



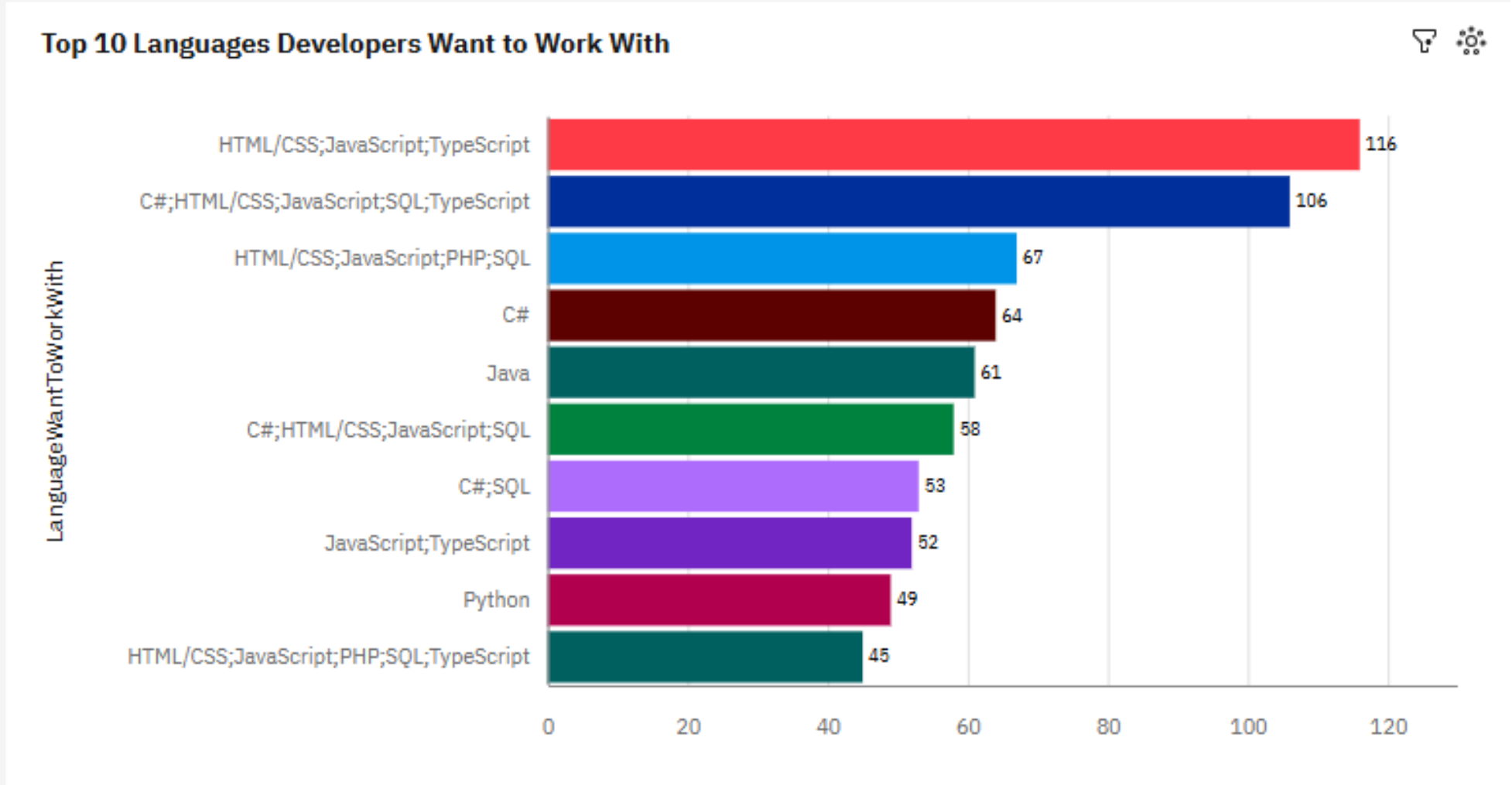
POPULAR LANGUAGES



PROGRAMMING LANGUAGE TRENDS



PROGRAMMING LANGUAGE TRENDS



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

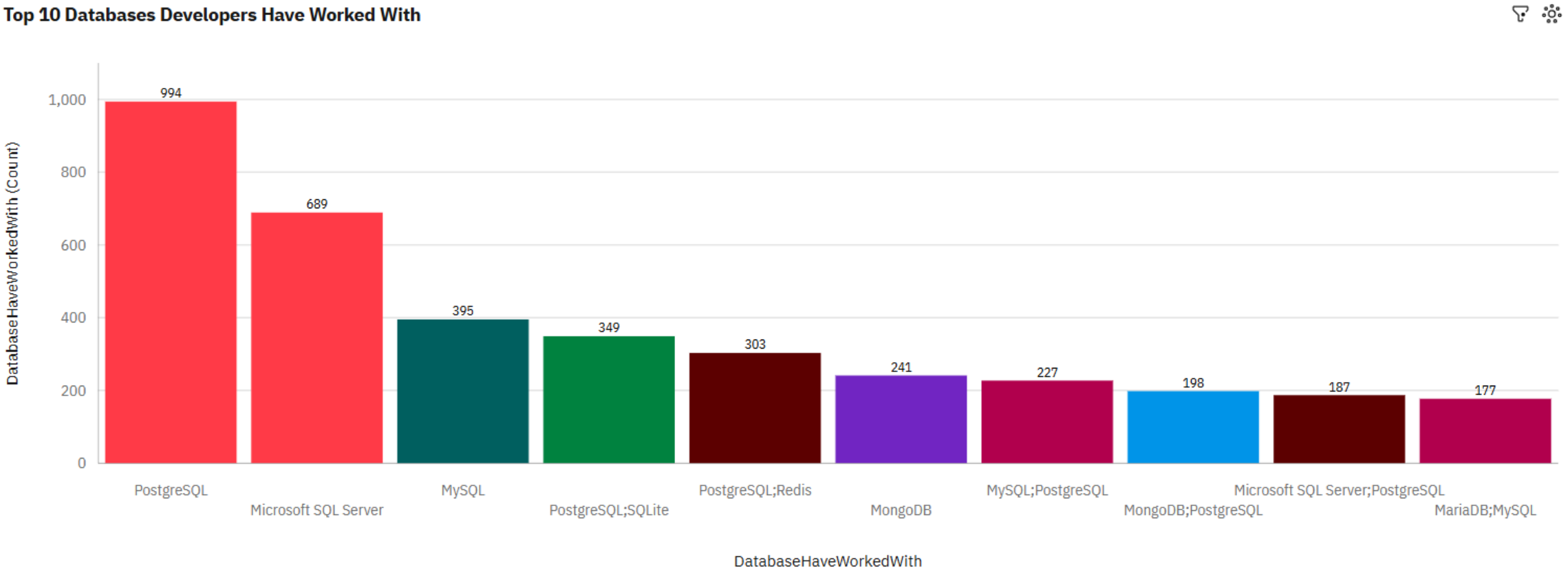
- **Full-stack web technologies dominate developer usage**, particularly combinations involving **HTML/CSS, JavaScript, and TypeScript**.
- **Established enterprise languages such as C# and Java remain highly relevant**, showing strong presence in both current and future trends.
- **Python continues to maintain strong demand**, reflecting its importance in data science, automation, and backend development.

Implications

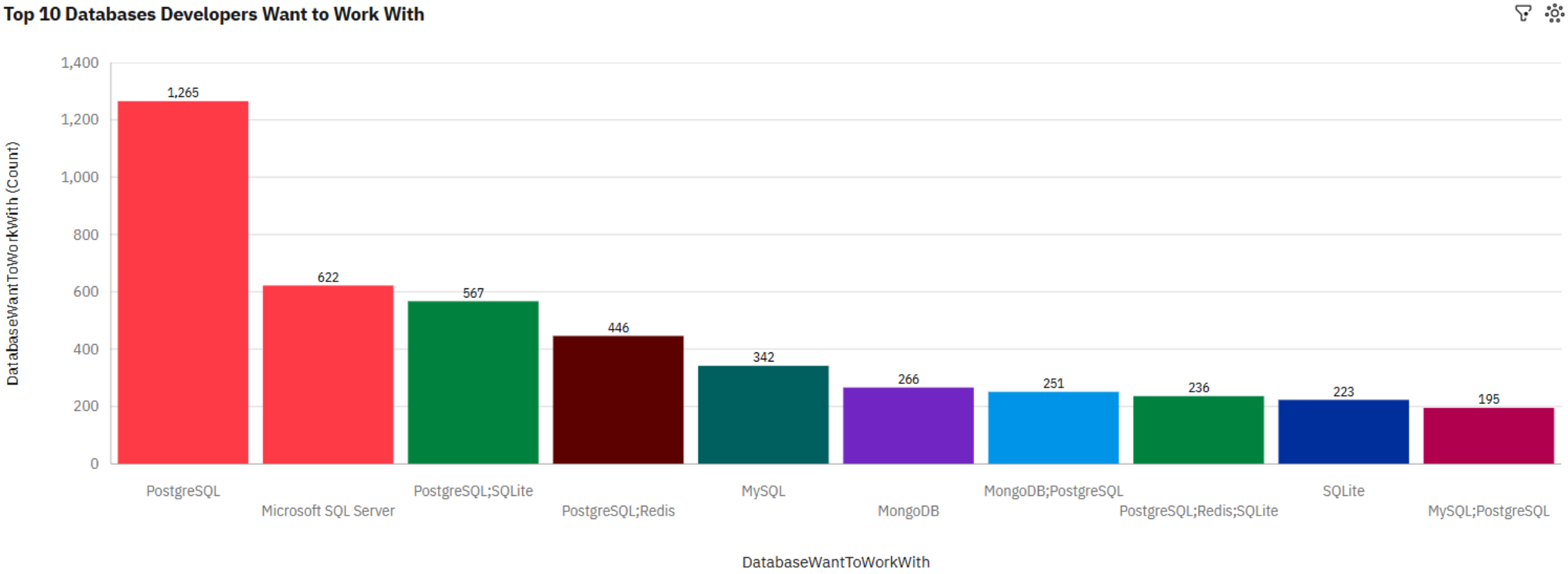
Developers should prioritize full-stack development skills, especially JavaScript frameworks and modern web technologies.

- **Organizations will continue to rely on enterprise backend languages**, meaning skills in **C#, Java, and SQL ecosystems** remain valuable.
- **Learning versatile languages like Python and JavaScript increases career opportunities**, as they are widely applicable across industries.

DATABASE TRENDS (Current Year)



DATABASE TRENDS (Next Year)



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- **PostgreSQL is the most popular database**, leading both current usage and future demand.
- **Relational databases remain dominant**, with PostgreSQL, MySQL, and Microsoft SQL Server widely used.
- **NoSQL databases such as MongoDB are gaining attention**, reflecting growing interest in flexible data storage solutions.

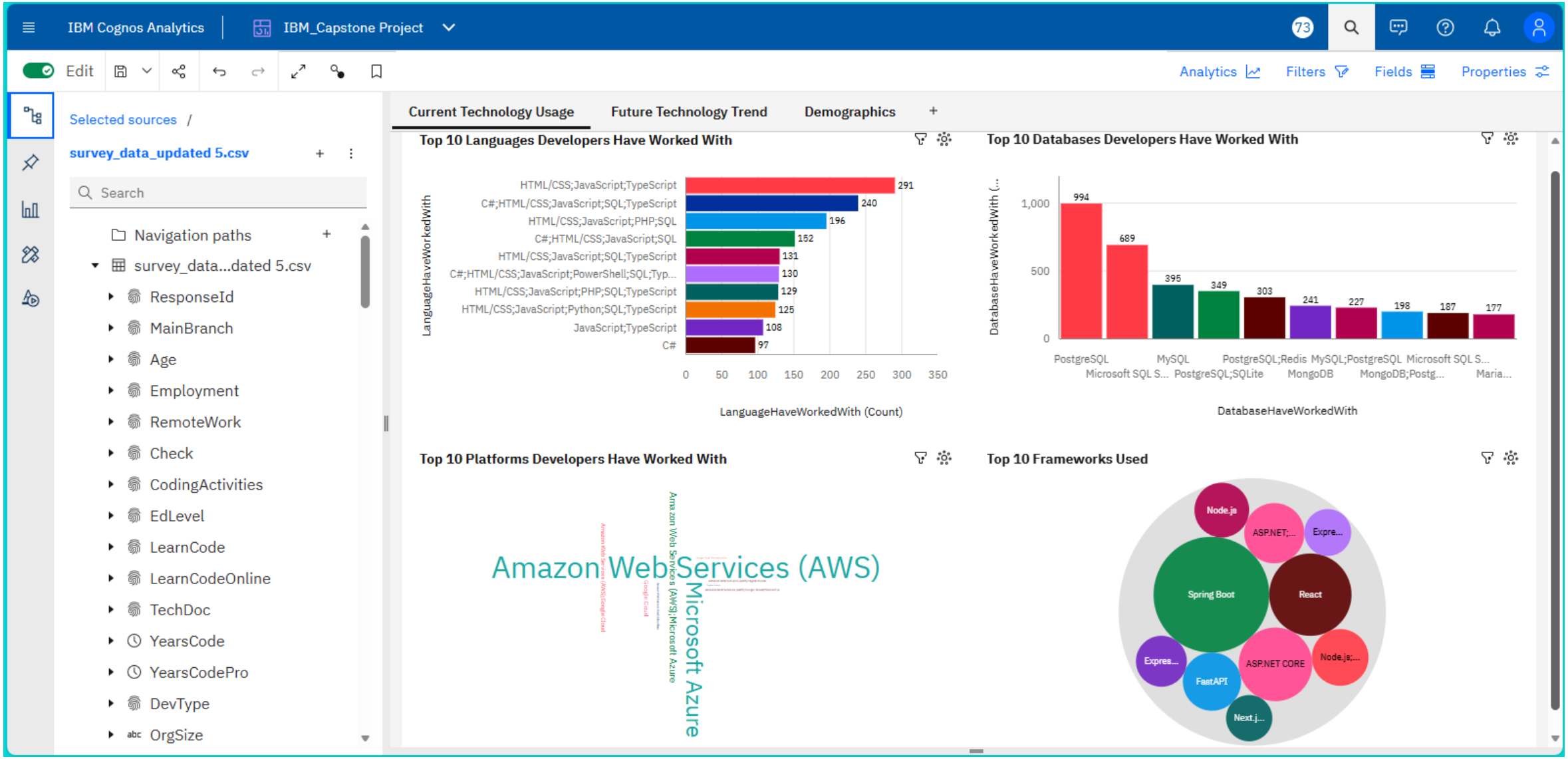
Implications

- **Developers should strengthen SQL and relational database skills**, particularly PostgreSQL.
- **Organizations will continue relying heavily on relational database systems**, especially for enterprise applications.
- **Knowledge of both SQL and NoSQL technologies increases employability**, as modern systems often combine both database types.

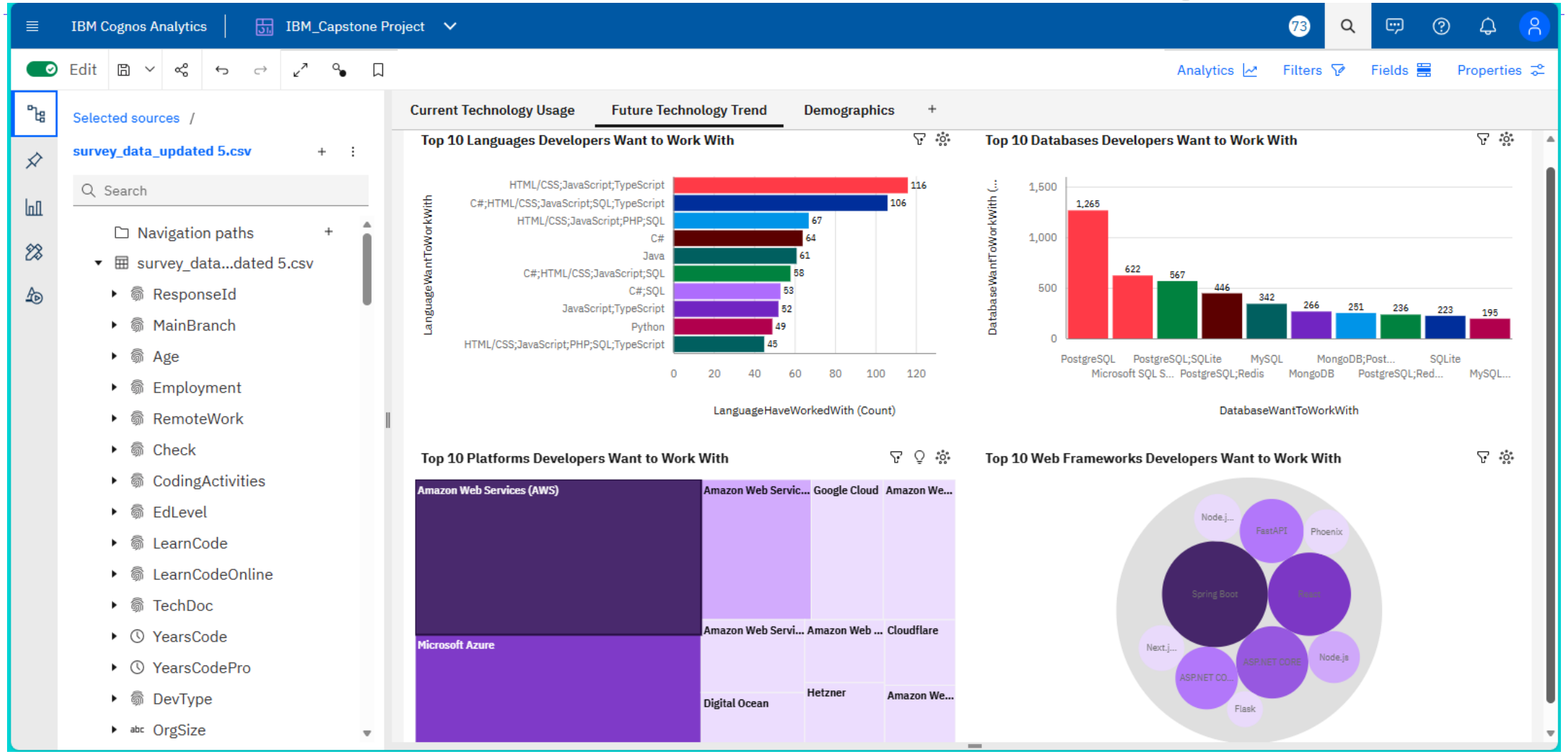
DASHBOARD

>>>

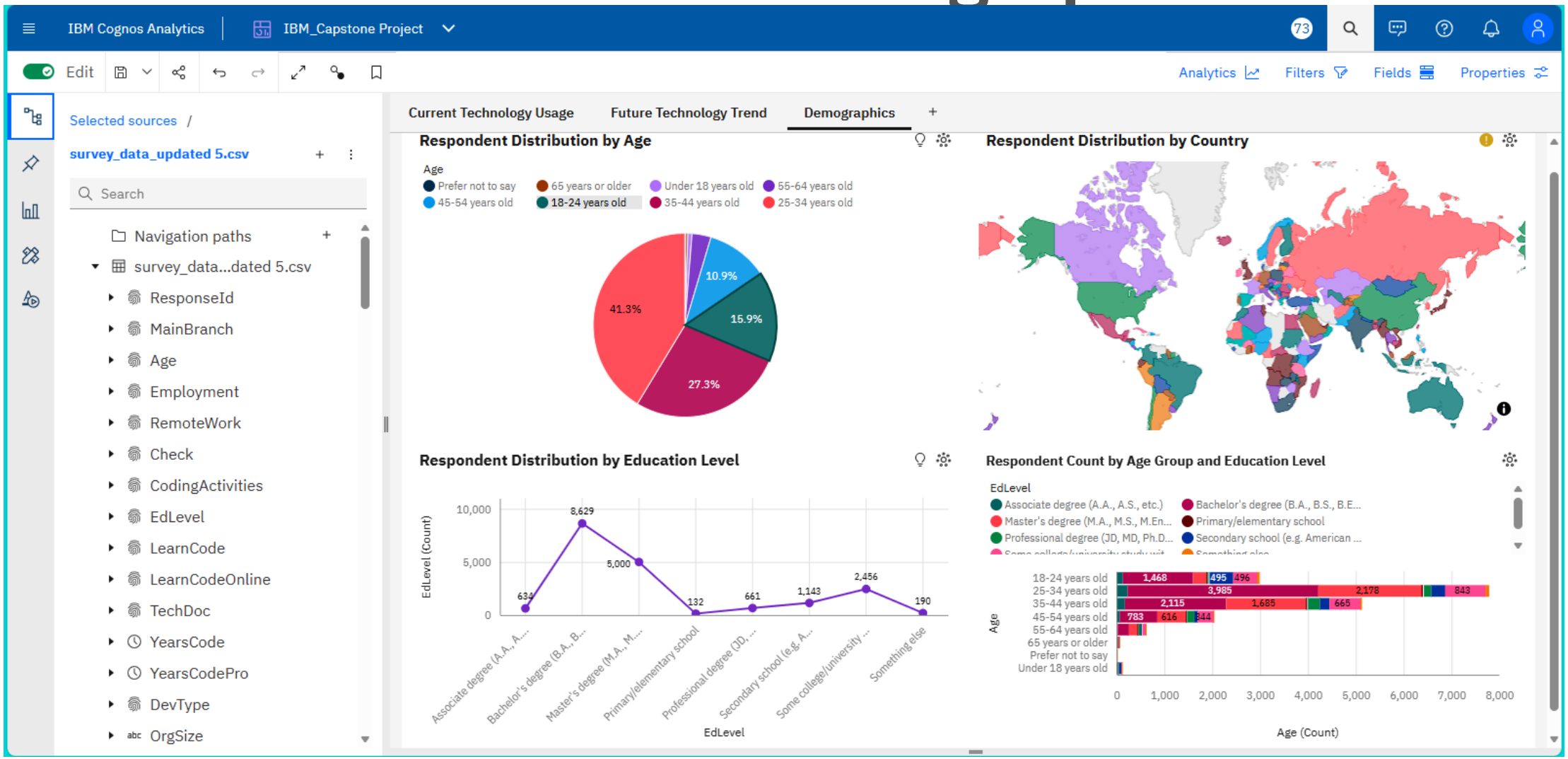
DASHBOARD TAB 1: Current Technology



DASHBOARD TAB 2: Future Technology Trends



DASHBOARD TAB 3: Demographics



DISCUSSION

The dashboard reveals a strong alignment between current usage and future demand in core technologies such as JavaScript, PostgreSQL, and cloud platforms. The demographic distribution indicates a young and globally diverse developer workforce.

The findings suggest that foundational technologies remain dominant, while cloud and modern frameworks continue to grow in importance.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Full-stack technologies dominate developer ecosystems.
- PostgreSQL is the leading and fastest-growing database.
- Cloud platforms represent the future direction of infrastructure.

Implications

- Developers should prioritize full-stack capabilities.
- Database optimization skills are critical.
- Cloud certification increases career competitiveness.

CONCLUSION

- Web technologies remain central to modern development.
- PostgreSQL shows strongest long-term demand.
- Cloud platforms continue to expand.
- Data-driven insights support strategic workforce planning.